

Lab: Real-Time Financial Data Pipeline (Kafka + Spark + HDFS + Superset)

Objective

Design and implement a **real-time financial analytics pipeline**:

- Ingest financial data from an API
- Stream via Kafka
- Process with Spark
- Store in HDFS (data lake)
- Visualize with Superset

Architecture Overview

Financial API => Spark (Producer) => Kafka => Spark (Consumer) => HDFS => Superset

Part 1 — Environment Setup

Student Tasks (Questions)

1. What are the roles of:
 - Kafka
 - Spark Streaming
 - HDFS
 - Superset
2. Install and start:
 - Kafka + Zookeeper
 - HDFS (single node or pseudo-distributed)
 - Spark
 - Superset
3. How do you verify that:
 - Kafka is running?
 - HDFS is accessible?
 - Spark is correctly configured?
4. Create a Kafka topic:
5. `financial_data`
 - What command is used?
 - How many partitions should you define and why?

Part 2 — Data Ingestion (Spark → Kafka)

Scenario

You will simulate or connect to a financial API (stocks, crypto, forex).

Student Tasks

1. Choose a financial API (or simulate data):
 - What fields are relevant? (price, volume, timestamp, symbol...)
2. Write a Spark job to:
 - Fetch data periodically
 - Convert it into JSON format
3. How do you send data to Kafka using Spark?
4. What is the structure of a Kafka message?
5. How do you ensure:
 - Fault tolerance?
 - Data consistency?

Part 3 — Stream Processing (Kafka → Spark)

Student Tasks

1. Create a Spark Structured Streaming job to:
 - Read from Kafka topic `financial_data`
2. What schema will you apply to the incoming JSON data?
3. Perform transformations:
 - Compute moving average price
 - Filter abnormal values
 - Aggregate per symbol
4. What is the difference between:
 - Batch processing
 - Structured streaming
5. How does Spark handle:
 - Late data?
 - Fault recovery?

Part 4 — Data Storage (Spark → HDFS)

Student Tasks

1. Store processed data in HDFS:
 - What file format will you use? (Parquet, ORC, CSV)
 - Why?
2. Design a **Medallion Architecture**:
 - Bronze (raw Kafka data)
 - Silver (cleaned data)
 - Gold (aggregated financial metrics)
3. How do you partition the data in HDFS?
 - By date?

- By symbol?
- 4. What are the advantages of:
 - Partitioning
 - Columnar formats

Part 5 — Kafka Streaming Behavior

Student Tasks

1. How does Kafka manage:
 - Offsets?
 - Consumer groups?
2. Simulate:
 - New incoming financial data
3. How does your pipeline react?
 - Is it real-time?
 - What is the latency?
4. What happens if:
 - Kafka broker crashes?
 - Spark job fails?

Part 6 — Visualization with Superset

Student Tasks

1. Connect Superset to your data:
 - How do you expose HDFS data? (Hive / Presto / Spark SQL)
2. Create datasets:
 - Financial time series
 - Aggregated metrics
3. Build dashboards:
 - Price evolution over time
 - Volume trends
 - Moving averages
4. What types of charts are best for:
 - Time series
 - Comparisons
 - Trends

Part 7 — Multidimensional Analysis (Advanced)

Student Tasks

1. Write queries to answer:
 - What is the average price per symbol per day?

- Which symbol has the highest volatility?
 - What are peak trading hours?
- 2. Use:
 - GROUP BY
 - ROLLUP
 - CUBE
- 3. How would you model:
 - Fact table (transactions/prices)
 - Dimensions (time, symbol, market)